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A Web and Mobile Based Approach to Redistribute Consumable Food Waste

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***Abstract*— This paper presents a web and mobile based approach to redistribute consumable waste foods. To increase the efficiency in the redistribution of consumable waste food, the main objective of this paper is to develop a web and mobile application to redistribute the waste food to the needy people in the society. In this project, a system is implemented that combines the topics of food surplus and scarcity of basic needs. Our proposed system achieves around 70% acceptance rate in UI and 75% in UIF. As the result, this designed system is reliable enough to satisfy the need for a centralized system to redistribute food waste. This system will help the needy people to get wastage food. Both mobile and web applications are user friendly and efficient.**

Keywords*— *Web Application, Surplus Food, Food Waste, Social Welfare, Leftover Food Management, Fight Hunger.

I. Introduction

Food and food waste has been an emerging global problem for the past few years, with 1 billion people malnourished problem globally.[14] Food can be wasted at different stage during harvest[15], food consumed[16], food for animals[17], or a set of social relations[18]. It is estimated that about one- third of the food is wasted [13] and this enormous amount of food can feed approximately 1.9 billion [19] people.

With accelerated economic growth and increasing production, food waste will be a matter of grave concern also in Bangladesh sooner or later [1]. Every seventh person in Bangladesh is an undernourished one who cannot acquire enough food to meet dietary energy requirements. Bangladesh has got 24.4 million undernourished people, or in other words 15.1 percent of its population is not having enough to meet their minimum daily diets[2], and on the other hand, the people in Bangladesh are wasting about 5.5% of the total procured food.[3] So, as we can see from those two studies, hunger is not a scarcity issue. There's more than enough food. It's actually a logistics issue.[4] After all, many restaurants, catering, and event companies often end up with plenty of leftovers, but don't want to cost or liability of figuring out how to donate and deliver them to those in need.[5] At the same time, those in need–not just the homeless, but also low-income families and the elderly–may not have the time or transportation to plug into existing free-meal services or food banks in their area. Similarly, we can say that it is true for other items like- clothes. Specially during winter poor people suffer from freezing cold for not having warm clothes, same situation is created during disaster and other ungodly hours. There's a gap between people in need of clothes and people who throw away wearable clothes.[6] There is a lack of centralized system to reduce food waste and redistribute among deprived people, which in result can reduce malnutrition, hunger, infant mortality rate and finally environmental impact.[7] In this project, a system is implemented that combines the topics of goods surplus and scarcity of basic needs. The project includes content designing, web portal development and mobile application development.

The key idea is to build a platform that will bring the people having surplus food and underprivileged people together by providing logistics support. Our proposed system will consist three parts: a) Donor, b) Volunteer, c) Recipient. By using web or mobile application donor will post an ad for donation providing necessary information, and nearby group of volunteers will be notified through the system. In next step, volunteers will collect the food from donor, and bring the donation to nearby processing center. In final step, the food item will be processed and make ready to serve at local processing center. Finally, processed food will be served to a group of needy people. An extension of our system is proposed for collecting donations from mobile balance for bearing logistic expenses.

II. Related Work

A. In Dhaka

'Prochesta Foundation' has been collecting foods and re-distributing them among the poor since May 2016 [8]. From its journey, this organization has helped many underprivileged people in Dhaka city by redistributing excess foods from wedding. Through this organization took a new project entitled 'Food Bank', and opened up a hotline so that their volunteers can be contacted for collecting food from the wedding venues. This organization runs based on Facebook page [9]. To donate, donor have to call their hotline number or have to contact b their Facebook page, and their volunteer will collect the leftover foods from that location. This organization functions within Dhaka only. In [21-30] mobile and web application have been developed for other applications. However, in this paper a web and mobile application has been developed to redistribute the consumable waste foods. The applications are very user friendly and efficient.

Phone Credit for Refugees

The Phone Credit For Refugees collects donation from donor's phone credits and distributes the donations to the refugees. Today, the group has over 64,000 members, has completed over 30,000 tops ups, and has raised over half a million pounds. Their support has expanded to cover refugees across Europe, Asia, Africa, and Middle East. [10] In summery this organization opens the door for small donation by transforming millions of small mobile credits into a large-scale donation. [11] However, this idea can be integrated into our project for managing organizational expenses from donation. In this time lots of organizations are working for solving this hunger problem

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And they are working in a successful way.[12] Food bank for NYC: This organization oversees feeding New York City's hungry, a big job for anyone. It was founded in 1983 as a response to the increasing number of people seeking emergency food assistance in New York City. The Food Bank For New York City works to end food poverty and increase access to affordable food for low-income New Yorkers throughout the five boroughs. Food for the hungry: This faith-based organization has worked since 1971 to end hunger globally, serving those in need in 20 countries worldwide. Instead of general donations, Food for the Hungry offers specific items that are in need with monetary amounts attached; for example, you can donate 20 food tree seedlings for $27, a sheep for $54, or buy the whole farm for $1,034. You can also sponsor a child for a monthly payment. Also, there are lots of such organization in Asia like serve needy, food for hunger, feed the poor.

III. Working Procedure

The Needy Serve: Web and Mobile application-based system will have three units (1) Recipients (underprivileged people), (2) Donor, (3) Volunteers/Agents. Mobile and web application will work as a platform to connect the donor and the volunteer. Both web and mobile application will share the same database to keep application integrity. Donor will post an ad of the item(s) he/she wants to donate through the application. The application stores the information into the database. From database, this information will be fetched and displayed onto other applications (accessed by other users) and also nearby registered volunteers will be notified through text messages to confirm if they can collect the item from donor. After the agent confirms his availability, donor will be informed about estimated collecting time as well as volunteer's information. Then collected item will be redistributed among needy people through local hub.

Donation(money) will be collected from users' mobile balance so that small amount of money can be collected in large number, and also to make the donation process easier. To collect donations through mobile balance, donor will be redirected to donate option for a particular mobile operator. By following mentioned options, donor make a donation through SMS. In return SMS, a pin number will be given to keep track of donation and specific amount will be deducted from balance.

IV. System Architecture

Our proposed system named Needy Serve is divided into three part: a) Mobile Application, b) Web Application, c) Database. Mobile and web application both share similar features and those features are divided into three units: Features for a) Donor, b) Volunteer, and c) Admin. For all users (Donor, Admin, and Volunteer) registration is must. User have to register first providing name, address, email and phone number. After successfully being registered, user can proceed to next steps.

A. Features for Donor

By default, every user is a donor. So, any user can access option related a) Donating leftover food, and b) Donating money from mobile balance, no special permission is required.

1. *Donate Left Over Food*

To donate surplus food, donor have to fill up a form mentioning the short details of food item(s), address of collecting the item, approximate servings (mention how many people can be feed with the food), and collecting time. For specifying "Collecting time", system will prompt the user to selection a duration (i.e.: 6 hours, 12 hours, 24 hours.). After a selection is made, system will make the donation ad available or valid for the mentioned duration from the time of the ad is registered. For example: if a donor selects 6 hours of duration time on 24-Jun-2019 at 3.00 PM (system time), system will register the donation ad valid till 24-Jun-2019 at 9.00 PM. After passed validation date, this donation ad will be automatically from system. Fig. 1 shows the flow chart for registering a donation ad to system.

On successful ad registration, the item will be available for nearby volunteer to collect.

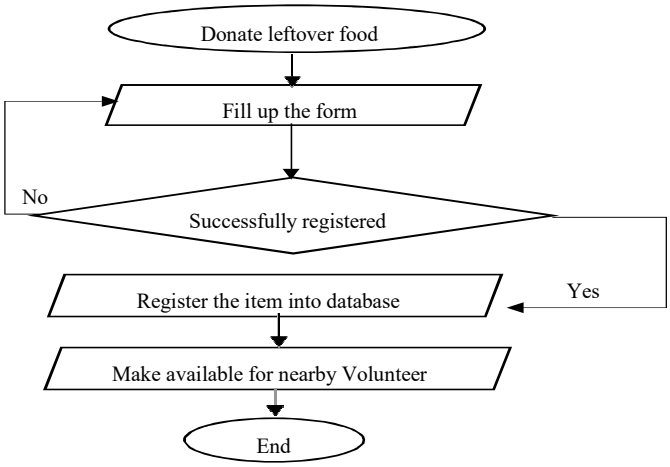

Figure 1: Flow chart for registering a donation Ad to system.

2. Donating money from mobile balance

For making a donation through mobile balance, user will be redirected to fill up a form. In this form, user have to mention a certain amount (Minimum 10 Tk to Maximum 100Tk), mobile operator. Upon fill the form, text message box will be opened to send a message for transferring the mentioned amount from user's mobile balance. If the amount is more than 100 Tk, donation will be collected through local mobile banking system (i.e.: BKash, Rocket).Finally, an admin will review the donation and it will be registered against the user.

B. Feature for Volunteer

To register as a volunteer user, first user has to place request. An admin will review the request, upon approval user can work as a volunteer.

After being approved as a volunteer, user will prompt to select his nearby area from where he can collect donations easily. Based on the area of a user, only the ads of that area will be displayed to the user. For example, if a volunteer selects Dhanmondi as his local area to serve, only the donations that are available to collect from Dhanmondi will be displayed to that volunteer. Volunteer have to select an ad for viewing the detail information of a particular donation. Finally, by approving, this particular donation item will be registered against the volunteer. Volunteer will collect the from the location and report back to the local hub. For security purpose, volunteer's contact details will be sent to the donor. Figs. 2 and 3 shows the block diagram of flow charts.

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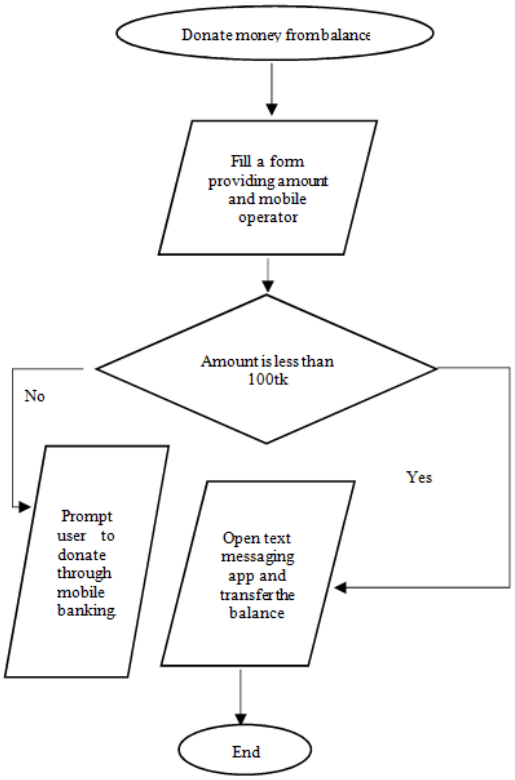

Figure 2: Flowchart for transferring donation from user to system.

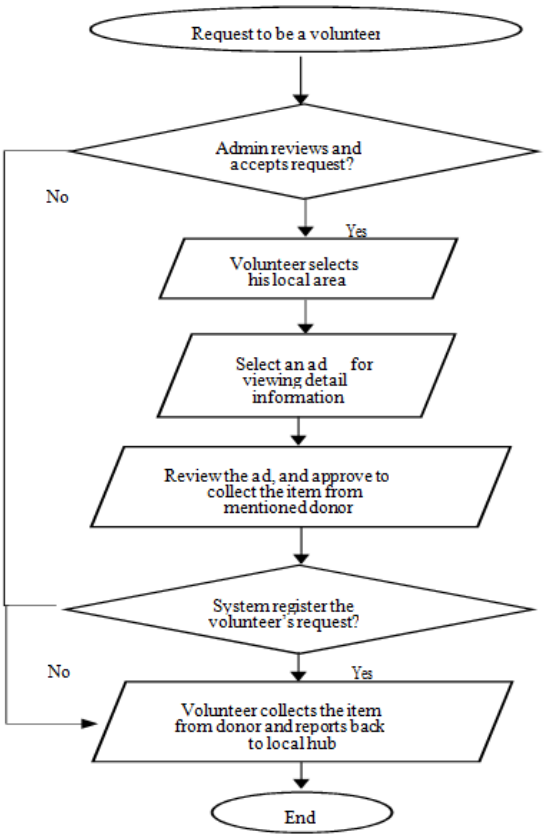

Figure 3: Flowchart of collecting donation from donor as a volunteer.

C. Features for Admin

Admin can also be described as system admin will performed the following task.

Clean database: After an ad expires, it is invisible to the users but still the information is in the database, which is nothing but a junk data. Those data are responsible for poor system performance to some extent. Admin has the right to delete invalid ads from database from app.

Accepting request of volunteer: Admin will review the volunteers' request and make an approval. After being approved by admin, a user can work as a volunteer.

Restrict access of users: Admin may restrict a particular user to use the system if security threat is raised or violates the code of conduct.

Though admin has the permission to modify database, he can't look up into the internal data structure. All operations will be performed by clicking buttons related to the operation.

USER INTERFACE

1. Web Application

A) Website Sitemap

A site map is a model of a website's content designed to help both users and search engines navigate the site. A site map can be a hierarchical list of pages (with links) organized by topic, an organization chart, or an XML document that provides instructions to search engine crawl bots. When the site map is for users, it's just a plain HTML file with a listing of all the major pages on a site. Fig. 4 shows the web application site map

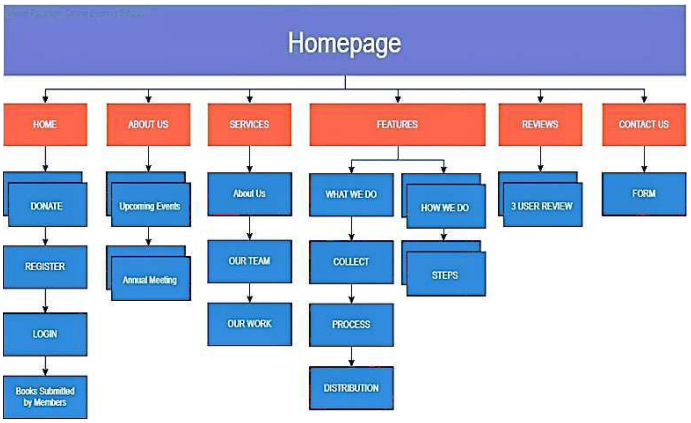

Figure 4: Web Application Sitemap 1.

A) Improvement Relationship Diagram

Relations Diagrams are drawn to show all the different relationships between factors, areas, or processes. These diagrams make it easy to pick out the factors in a situation which are the ones which are driving many of the other symptoms or factors. Instead of one item following another in a logical sequence, each item is connected to many other pieces, showing that they have an impact on each one. Once all the relevant connections between items have been drawn, the connections are counted. Those with the mostconnections will usually be the most important factors to focus on.

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Fig. 5 shows the improvement relationship diagram. Fig. 6 shows home page of the website for the proposed system. Fig. 7 shows user profile interface.

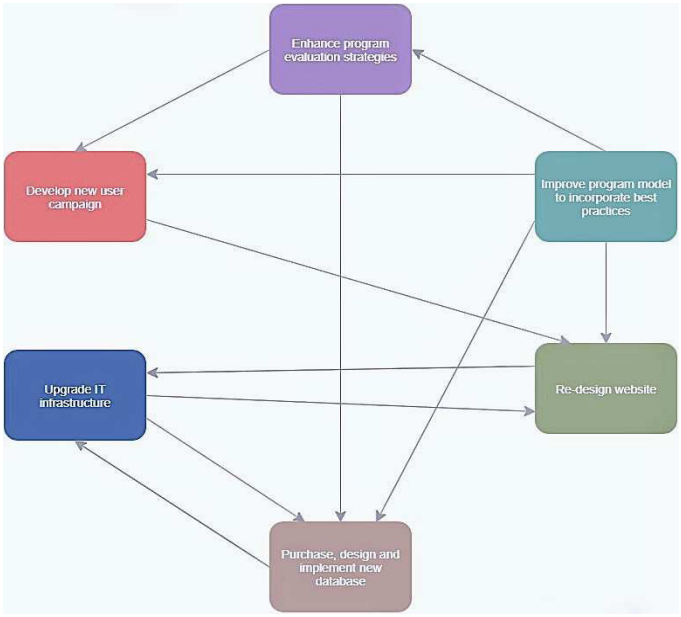

Figure 5: Improvement Relationship Diagram

A. Website Designed Prototype

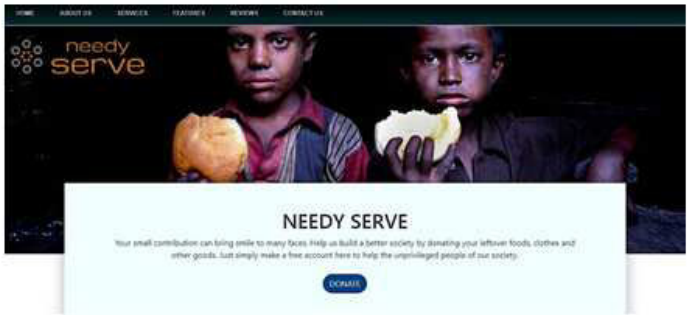

Figure 6: Home Page

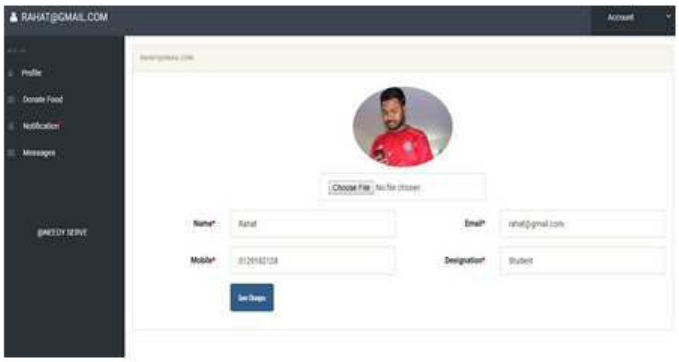

Figure 7: User Profile Interface

A) Mobile Application Interface

My Menu

Both user and volunteer have to register into the system before gaining the access. Only basic information is required for completing registration. To make the registration process hassle free and more user friendly only basic information is considered for a user. User have to provide his Name, Email address, Phone number, Password that will be used for verification in login purpose, and Gender. Each of the fields are non- empty. For gaining access to the system each of the information have to be provided. Among the provided information email address should have to be unique, that means only one user can be registered against one email address. If there is any other user registered with a particular email address, then that email address can't be used for another user. Unique email address is assigned to users for keeping system integrity. Moreover, email address is used for user verification, and in that case if an email address is assigned to multiple user then there is a risk for security. Any user may gain access to multiple account and its information vice-versa. Here, users phone number may be seemed unnecessary, but phone number is used for ensuring rapid communication between Donor and Volunteer. As to this phase, our application doesn't use map or similar feature to point exact location, donors' and volunteers' mobile number will help to build a rapid communication among the audiences. Password is for verifying the user while login, and Gender is kept for ensuring equal participation of the users. Finally, a register button is at the bottom of the page to make a connection to the database for registering the user into the database. Upon clicking the Register button, a background thread collects the input form the page, pass these inputs into background thread and the thread handles the operation consists of making a connection with the database, pass the parameter to the database and on completing successful execution finally collects the response from the database. Base on the response from the database, system decides if a user is successfully registered or not. Fig. 8 shows the user profile interface of mobile application

Figure 8: User Profile Interface of Mobile Application

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After successful, login user is redirected to a Navigation Drawer page. This page contains, navigation menu which can be accessed from upper left corner by clicking three parallel white bars. Initially, landing screen consists of the posted ad of different donors. Latest add from donors is showed first. These ad are designed in a titles manner, and contain all information regarding a posted food item for donation including name of the donor, location of the pickup point, collect time (Donor have to define the expiry date of the item), Approximate Serving (How many people can be served with this food, Food description (A short description of food including what is used to cook and the current condition of the food that means if the food is Biryani, then the preferred description is ingredients are: meat, rice description would contain preserved well etc).

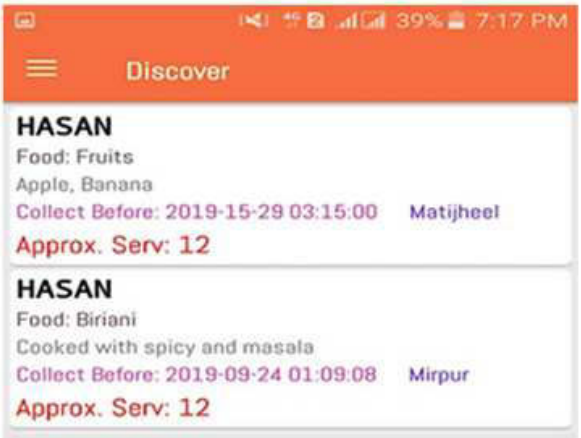

Figure 9: Discover Layout listing Food item for donation.

A) Feature for Volunteer

To register as a volunteer user, first user has to place request. An admin will review the request, upon approval user can work as a volunteer. After being approved as a volunteer, user will prompt to select his nearby area from where he can collect donations easily. Based on the area of a user, only the ads of that area will be displayed to the user. For example, if a volunteer selects Dhanmondi as his local area to serve, only the donations that are available to collect from Dhanmondi will be displayed to that volunteer. Volunteer have to select an ad for viewing the detail information of a particular donation. Finally, by approving, this particular donation item will be registered against the volunteer. Volunteer will collect the from the location and report back to the local hub. For security purpose, volunteer's contact detail will be sent to the donor by SMS.

V. Testing

To reduce system fault in every stage, fault removal is implemented in developing every version of mobile application. Fault removal consists of checking and analysis process which are requirements reviews, design reviews, code inspections, and product testing.

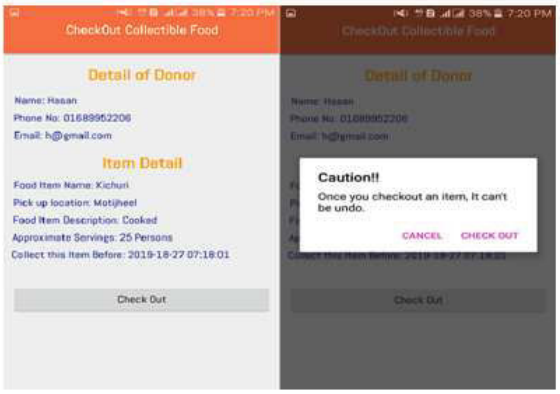

Figure 10: Checkout process for Volunteer.

Following tests have been conducted in testing our designed system.

Integration Test – Take multiple individual unit at a time and make sure every unit interacts perfectly.

Load Test – Load test includes testing with field data with accounted interactions.

Regression Test – Check if a major change in code causes a negative effect in the functionality of the application. **Acceptance test –** Test conducted in participation with user if system meets the acceptance rate.

For conducting acceptance test on our designed system version 4.0, a total of 43 users was selected including faculty, stuffs and students as a participant. The purpose of the acceptance testing is to check if the designed system meets the functional and non-functional expectations in the requirements determination phase and if the system matches exactly or comes closely matching users' expectation [20]. The whole test was focused on three areas in total a) basic functionalities, b) UI (user interface), and c) UFI (user friendly navigation). It is important that at least 70% of the agrees that our designed model meets their expectation. Acceptance test concludes that 85% users agreed that the system has their expected basic functionality as a leftover food distribution application. 75% user believes that system is having a user-friendly navigation, and 73% user believes that system's user interface meets their expectation.

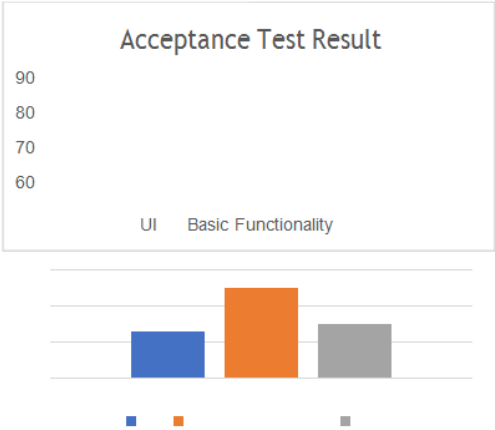

Figure 11: Acceptance test rate (%) in different categories.

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As every category is above the pass rate 70%, we may consider that our designed system has passed the acceptance test. However, any feedback from users will help to improve the rate in those categories in future.

VI. Conclusion and Future work

Needy Serve, our designed system meets all functional and non-functional requirements. As the Needy Serve is built upon well-analyzed and well-studied business model, it meets the user requirements in every level. Needy Serve ensures more efficient redistribution process of leftover food for underprivileged people with minimum logistic cost. It can be concluded that Needy Serve will fight the hunger problem more efficiently by reducing the food waste.

Though the system has passed the acceptance test, acquired rate in most of the categories is good but not best. Acceptance score in UI is 70% and in UIF is 75%, which indicates that system's user interface design is not the best that can be achieved. To remove the barrier between designer and user, user can be asked for suggestion and feedback regarding application's user interface design. By implementing the suggestion and feedback from users, system acceptance rate regarding UI and UIF can be dramatically improved.

We have only worked with leftover food, but there are other necessary items like cloth, which can be redistributed specially during winter and rainy season. Thus, we don't need to organize any campaign during winter to collect warm clothes for needy people. Donor will simply put an ad like food item, and our volunteer will collect the item.

One major setback in collection of funds as a donation for bearing logistic expenses is that we are now transferring balance from users' mobile to our system's mobile number. This transfer isn't free but comes with a service charge. Sometime this service charge takes 25% of the donation if the amount is less than 25 TK. So, it can be concluded that this service charge for collecting mobile donation from users' mobile balance is doing nothing but increasing the total expense. This dilemma can be resolved if mobile operator provide us with an API for collecting donation from mobile balance with free of charge.

This system is designed provided that a group of people is ready to join as a volunteer, but this situation can't be guaranteed. Various campaigns can be organized to motivate people for participating in social work to help needy people in our society.

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